

TANMAY SINGH

+91 8860028629 | tanmaysingh8970@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Bengaluru, Karnataka, India
[GeeksforGeeks](#) | [HackerRank](#) | [LeetCode](#)

Summary

Final-year B.E. student in Artificial Intelligence and Data Science who builds AI software that solves practical problems. Skilled in Python, machine learning, deep learning, computer vision, and web development. Built a hand-gesture game controller that recognizes gestures with 99% accuracy, a self-learning agent that plays a game to find bugs on its own, and an AI shopping assistant that completes secure test payments. Seeking an entry-level AI Engineer role.

Education

K. S. School of Engineering and Management

Bengaluru, India

Bachelor of Engineering in Artificial Intelligence & Data Science | CGPA: 7.3/10

2023 - 2027

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML, CSS

AI & ML: PyTorch, scikit-learn, Stable-Baselines3, Gymnasium, OpenCV, NumPy, Matplotlib, Google Gemini API

Frameworks & Libraries: Next.js, React, Flask, Pygame

APIs & Databases: REST APIs, Razorpay API, PostgreSQL

Developer Tools: Git, GitHub, GitHub Actions, Docker, Vercel, CUDA

Projects

CNN-Based Gesture Controlled Pac-Man | *Python, PyTorch, OpenCV, CUDA, Pygame*

[View on GitHub](#)

- Built and trained a MobileNetV2-based Convolutional Neural Network (CNN) for real-time hand gesture classification using a 2,000-image HaGRID dataset, achieving 99.0% test accuracy across left, right, up, and down commands
- Developed a real-time computer vision pipeline using OpenCV, CUDA-enabled PyTorch inference, confidence thresholding, and temporal smoothing to convert webcam gestures into responsive game controls
- Engineered a Pac-Man-style Pygame application with maze navigation, ghost AI, pellets, power-ups, scoring, lives, and level progression, maintaining approximately 60 FPS gameplay with 30 FPS gesture recognition

Glitch Hunter - AI Game Testing System | *Python, Stable-Baselines3, Gymnasium, Flask*

[View on GitHub](#)

- Trained a PPO reinforcement learning agent in two stages for 16 million steps across 8 parallel environments to autonomously explore and interact with a Mario-style Pygame game for continuous gameplay testing
- Developed a runtime glitch detection system that monitors game-state invariants to identify abnormal movement, clipping, score, and coin-related bugs, saving screenshot, GIF, and PDF report evidence for each bug
- Built a live Flask and Socket.IO dashboard to stream AI gameplay, actions, rewards, and detected glitches for real-time testing and analysis

Razorpay Agentic Commerce | *TypeScript, Next.js, PostgreSQL, Gemini API, Razorpay API*

[View on GitHub](#)

- Built and deployed on Vercel an AI buyer agent using Google Gemini to understand shopping requests and propose suitable products within user-defined requirements
- Developed a secure policy and human-approval system where the server validates prices, spending limits, inventory, and authorization before payment
- Integrated Razorpay Test Mode with payment state tracking, retries, inventory reservation, and audit logging for reliable, complete purchases

Hackathons

Hire-4-Thon - National Level Hackathon (2026)

[View Certificate](#)

Hackathon-24 - College Level Hackathon (2024)

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Certifications

Introduction to Machine Learning - VOIS (2025)

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Getting Started with Artificial Intelligence - IBM SkillsBuild (2024)

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Generative AI Literacy - IT-ITeS SSC / FutureSkills Prime (2025)

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